Curated Research Articles

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- **Guiding sulfur without crystallizing it** — score: 1.000 The article details a study that utilizes real-time operando techniques to reveal how controlled disorder in sulfurized polyacrylonitrile enhances the stability of short sulfur chains. This innovation not only reduces polysulfide loss but also significantly improves the reversibility of lithium-sulfur batteries. Journal: *Nature Materials*
- **Observing energy materials in action** — score: 1.000 The article discusses the significance of operando measurements in studying energy materials, which allow researchers to observe intricate electrochemical processes in real-time and gain detailed understanding of their mechanisms. This approach enhances the analysis of energy materials' performance and behavior under actual operational conditions. Journal: *Nature Materials*
- **Deciphering the Failure Mechanism and Improvement Strategy of High-Sulfur-Loading Na-S Batteries and Promoting Polysulfide Conversion by Electrochemically Reconstructed Cu1.84Mo6S8/Na2Cu4S3 Heterostructure** — score: 1.000 The article investigates the failure mechanisms of high-sulfur-loading room-temperature sodium-sulfur (Na-S) batteries and presents a dual-level strategy to enhance their performance. By designing innovative separators and employing an electrochemically reconstructed Cu1.84Mo6S8/Na2Cu4S3 heterojunction, the authors demonstrate improved conversion of sodium polysulfides (NaPSs) and increased catalytic efficiency, resulting in enhanced electrochemical performance and cycle life for the batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A Quantitative Lithium Inventory Framework for Anode-Free Lithium Metal Batteries** — score: 1.000 This article presents a quantitative framework for tracking lithium inventory in anode-free lithium metal batteries, specifically examining NMC622||Cu pouch cells. It identifies three key degradation stages during battery cycling that affect performance—Li depletion from the cathode, accumulation of inactive lithium, and thickening of the solid electrolyte interphase (SEI)—emphasizing the critical role of the cathode's stability and lithium buffering capacity for enhancing cycle life and safety in these batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Breaking the Electronic Reactivity and Ionic Mobility Trade-off via Confined Water Redistribution for Hydroxide-ion Storage** — score: 1.000 The study presents a novel approach to enhance both electronic reactivity and ionic mobility in -Ni(OH)2 electrodes for Ni-Fe batteries by manipulating confined water via annealing. This method boosts electronic conductivity by broadening the eg* band through increased coordinated water while optimizing ion mobility by reorganizing hydrogen-bonding networks, resulting in improved storage capacity (321.8 mAh g−1) and enhanced cycling stability (48% increase). Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Engineering Spatiotemporally-Resolved Hydrogen-Bond Networks for Decoupled H+/Zn2+ Transport Toward Durable Low-Temperature Aqueous Zinc-Ion Batteries** — score: 1.000 The article discusses a novel approach for enhancing low-temperature aqueous zinc-ion batteries by engineering a hydrated deep eutectic electrolyte (HDEE) that optimally regulates hydrogen-bond networks. This strategy decouples the transport of protons and zinc ions, thereby reducing unwanted side reactions and promoting even zinc deposition, resulting in significant improvements in battery performance and longevity at low temperatures. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Identification of Lithiation Mechanisms in Silicon-Carbon Multilayer Electrodes for Li-Ion Batteries Using Operando Neutron Reflectometry** — score: 1.000 The article discusses a study that employs operando neutron reflectometry to investigate the lithiation mechanisms occurring in silicon-carbon multilayer electrodes used in lithium-ion batteries. This research aims to enhance understanding of how these electrodes perform during battery operation, potentially leading to improved battery efficiency and longevity. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*

- **[ASAP] Structure–Transport Relationships in Microarchitected LiFePO4–Carbon Li Ion Battery Electrodes** — score: 0.800 The article explores the correlation between the microarchitectural structure of LiFePO4–Carbon electrodes and their transport properties in Li-ion batteries. It provides insights into how these structural features influence the performance and efficiency of battery electrodes. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Facing X-ray Radiation Damage Systematically: A Quantitative Investigation on Solid-State Electrolytes for LiS Batteries** — score: 0.800 This article presents a detailed quantitative analysis of solid-state electrolytes used in lithium-sulfur (LiS) batteries, specifically focusing on their behavior and resilience when exposed to X-ray radiation damage. The study aims to enhance the understanding of material performance under radiation conditions, which is crucial for the development of more durable battery technologies. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **Dynamic Cross-Linking of Organosulfur and Organoselenium in a Biphasic Cathode for Balanced High-Performance Lithium Batteries** — score: 0.800 The article details the development of a biphasic organochalcogen cathode that combines an organosulfur polymer with a liquid organoselenium compound through in situ cross-linking. This innovative approach enhances charge transfer and ionic diffusion, resulting in a significant reduction of performance trade-offs and achieving impressive metrics such as ultra-long cycling stability (over 2,700 cycles) and high areal capacity (11.36 mAh cm−2), positioning it as a promising solution for next-generation high-performance lithium batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Micro-Hydrophobic Microstructure Balance Electric Field Distribution and Wettability for Ultra-Stable Zinc Anodes** — score: 0.800 The article discusses the development of micro-hydrophobic microstructures on zinc anodes using photolithography techniques, which effectively balance electric field distribution and surface wettability, leading to the inhibition of harmful zinc dendrite growth and side reactions in aqueous zinc-ion batteries. The study identifies optimal surface configurations, particularly the Zn100° anode, which demonstrates remarkable stability and capacity retention over extensive cycling, marking a significant advancement in zinc anode technology and offering potential solutions for other metal anodes as well. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Pushing Energy Density Limits: Anode-Less Solid-State Battery Integrated With Amorphous Lithium Vanadium Oxide Thin-Film Cathode** — score: 0.800 The article presents advancements in anode-less thin-film solid-state batteries by utilizing a Sn/C seeded current collector, which enhances lithium plating stability and reduces nucleation overpotential. The improved design enables high energy density, achieving 235.13 Wh L−1 after 100 cycles, paving the way for more efficient microbattery systems tailored for integrated applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] The Decisive Role of Na Metal Anode Interfacial Chemistry in Governing the Na-Storage Behavior in Na//Hard Carbon Half-Cells** — score: 0.800 This article investigates how the interfacial chemistry of sodium (Na) metal anodes affects sodium storage behaviors in Na//hard carbon half-cells. The findings highlight that the reactions occurring at the Na anode interface significantly influence the overall performance and stability of sodium-ion batteries, emphasizing the need for careful consideration of interfacial properties in battery design. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Anti-Perovskite Nitride Decorated Bowl-Shaped Carbon: Advanced Hosts for Stable Lithium Metal Batteries at Low N/P Ratio** — score: 0.800 This study introduces an innovative bowl-shaped carbon (BC) host integrated with lithiopholic anti-perovskite nitrides, specifically assessing the Ni3ZnN/BC system. The architecture enhances lithium accommodation and volume control, achieving stable cycling in symmetric cells and an impressive performance in full cells with a low N/P ratio. This approach offers a promising framework for designing advanced hosts for lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Engineering disorder: Entropy as a design lever for stable interfaces in solid-state sodium-metal batteries** — score: 0.800 The article discusses the innovative approach of utilizing entropy as a design strategy to enhance the stability of interfaces in solid-state sodium-metal batteries. By engineering disorder in the battery's materials, the authors aim to improve performance and longevity, addressing key challenges in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Triple-phase boundary instability as a key degradation factor in sulfide|(oxy)halide dual-electrolyte solid-state batteries** — score: 0.700 The article investigates the instability at the interface between the sulfide electrolyte separator and the (oxy)halide electrolyte in dual-electrolyte solid-state batteries. By examining the degradation mechanisms influenced by different (oxy)halide chemistries, the research highlights the importance of this interface in affecting battery performance and longevity. Journal: *Joule*
- **Intergranular Degradations of Secondary NMC Particles in Liquid and Solid-State Environments** — score: 0.700 The article examines the mechanical degradation occurring at the intergranular level of secondary NMC particles used in lithium-ion batteries, highlighting its implications for enhancing cathode performance and reliability. It addresses the challenges posed by such degradations in both liquid and solid-state environments, contributing to the ongoing quest for improved energy storage solutions. Journal: *RSC - EES Batteries latest articles*
- **Spatial reconfiguration chemistry to boost closed-pore formation in polymer-derived hard carbon for superior sodium storage** — score: 0.700 The article discusses a novel approach in spatial reconfiguration chemistry aimed at enhancing the formation of closed pores in polymer-derived hard carbon. This advancement is significant for improving sodium storage capacity, with implications for the development of more efficient energy storage materials. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Reaction-Transport Coupling Driven by Supramolecular Host-Guest Chemistry Accelerates Kinetics in Solid-State Lithium-Sulfur Batteries** — score: 0.600 The article discusses advancements in solid-state lithium-sulfur batteries, highlighting how the integration of supramolecular host-guest chemistry enhances reaction-transport coupling and accelerates reaction kinetics. This innovation could potentially lead to improved battery performance and efficiency. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **A 50 µm-Thick Fiber-Reinforced Fluorinated Polymer Plastic Crystal Electrolyte Enabling High Ionic Conductivity and Voltage Stability for Advanced Solid-State Sodium Batteries** — score: 0.600 The article presents a novel 50 µm-thick fiber-reinforced fluorinated polymer plastic crystal electrolyte that significantly enhances the ionic conductivity, voltage stability, and mechanical strength of solid-state sodium metal batteries. By incorporating fluorinated polymers, the design optimizes Na+ transport dynamics and electrochemical stability, achieving impressive performance metrics and durability when coupled with various cathode materials, thus demonstrating its potential for advanced energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Time-Space-Medium Collaboratively Modulated LIBS Combined with Matrix Dilution: From Self-Absorption to Isotope Discrimination of Lithium** — score: 0.600 The article discusses a novel approach to Laser-Induced Breakdown Spectroscopy (LIBS) that utilizes collaborative modulation of time, space, and medium, along with matrix dilution, to enhance the detection of lithium. This method effectively addresses issues of self-absorption and enables precise discrimination of lithium isotopes, improving analytical capabilities in materials science and geochemistry. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Multifunctional Chloride-Oxide Additive Enabling Simultaneous Ionic/Electronic Conduction for High-Rate All-Solid-State NCM Cathodes** — score: 0.600 The article presents a multifunctional chloride-oxide additive that improves the performance of all-solid-state nickel-cobalt-manganese (NCM) cathodes by enabling simultaneous ionic and electronic conduction. This advance-

ment is crucial for enhancing the charge/discharge rates in high-performance energy storage applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*

- **[ASAP] Local Thermal Strain Regulated Solid Electrolyte Interphase with Advanced High-Temperature Tolerance** — score: 0.500 The article discusses the development of a solid electrolyte interphase (SEI) that is locally regulated by thermal strain, enhancing the SEI's tolerance to high temperatures. This innovation aims to improve the performance and stability of solid-state batteries, making them more viable for practical applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Streamlined V3.5+ Electrolyte Production by Leveraging Chemical and Catalytic Reductions** — score: 0.500 The article discusses a novel catalytic approach for producing V3.5+ electrolyte used in vanadium redox flow batteries, which enhances the efficiency of the process by integrating catalysis into the critical rate-limiting step. By doing so, the authors demonstrate a 67% reduction in production time and mitigate issues related to residual oxalic acid, while maintaining catalyst stability over extensive cycling, indicating significant improvements in the cost-effectiveness and performance of electrolyte production. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Oxygen Reduction Activity of High-Valence Vanadium in Synergy with N-Doped Carbon: A Nonprecious Metal Electrocatalyst for Zinc–Air Battery Cathodes** — score: 0.500 This article explores the development of a nonprecious metal electrocatalyst for zinc-air batteries, highlighting the enhanced oxygen reduction activity of high-valence vanadium when combined with N-doped carbon. The findings suggest that this synergistic effect can improve the efficiency and performance of cathodes, providing a promising alternative to expensive precious metal catalysts. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] Compositionally Complex Doping Enables High-Voltage Spinel Cathodes with Ultrafast Charging and Chemo-Electro-Mechanical Stability** — score: 0.400 The article discusses advancements in spinel cathodes for high-voltage applications achieved through compositionally complex doping. This innovative approach not only enhances the ultrafast charging capabilities but also improves the chemo-electro-mechanical stability of the cathodes, indicating significant potential for energy storage technology. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Molecular skeleton programming of premediators in sulfur electrochemistry** — score: 0.400 The article discusses the utility of 2-chloropyrimidine as a promising premediator for enhancing the performance of lithium-sulfur batteries through molecular skeleton programming. This approach not only improves capacity retention but also paves the way for the development of functional molecules across a wider range of organic chemistry applications. Journal: *Nature*
- **Navigating interfaces in solid-state sodium batteries: challenges, solutions, and future directions** — score: 0.400 The article reviews the advancements and challenges in solid-state sodium batteries (SSSBs), highlighting their potential as a cost-effective alternative to lithium-ion batteries due to sodium's abundance and improved energy density. It addresses the significant interface-related issues that hinder their commercialization and discusses possible solutions and future research directions. Journal: *RSC - Energy Environ. Sci. latest articles*
- **[ASAP] Sulfur-Doped Li2OHCl Antiperovskite Electrolyte for Stabilizing the Solid Electrolyte Interphase in Solid-State Lithium-Metal Batteries** — score: 0.400 The article discusses the development of a sulfur-doped Li2OHCl antiperovskite electrolyte aimed at enhancing the stability of the solid electrolyte interphase (SEI) in solid-state lithium-metal batteries. This advancement is significant for improving battery performance and longevity by addressing common issues related to SEI formation. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Molecular-Scale Hybrid Strategy Synergizes Solvation Regulation and SO42– -Derived Inorganic-Rich SEI for Zn Anodes** — score: 0.400 This article discusses a novel molecular-scale hybrid strategy that enhances the performance of zinc anodes by optimizing solvation processes and creating a sulfate-derived inorganic-rich solid-electrolyte interphase (SEI). This approach

aims to improve the efficiency and longevity of zinc batteries, addressing common challenges in their operation. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*

- **[ASAP] Bipolar Redox Behavior of Polyaniline Cathodes in Rechargeable Lithium Batteries** — score: 0.400 The article investigates the bipolar redox behavior of polyaniline when used as a cathode material in rechargeable lithium batteries. The study highlights the electrochemical performance and potential advantages of polyaniline in enhancing battery efficiency and lifespan. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Strain-Directed Ru Redistribution to Form RuO2/Pt Mosaic Heterointerfaces for Acid-Stable Water Oxidation** — score: 0.400 The study introduces a novel method for enhancing the stability of RuO2 in acidic environments by leveraging strain-driven atomic migration to create RuO2/Pt mosaic heterointerfaces within a PtNi framework. This innovative approach not only stabilizes the Ru4+ state and minimizes lattice oxygen participation during water oxidation but also results in a catalyst that exhibits impressive performance with an ultralow overpotential and exceptional operational stability, highlighting the potential of strain-directed atomic migration for advancing electrocatalyst design. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A donor-acceptor covalent organic framework as bipolar cathode for advanced sodium-ion batteries** — score: 0.400 The article discusses the development of a donor-acceptor covalent organic framework intended for use as a bipolar cathode in sodium-ion batteries. The authors highlight its potential to enhance energy storage performance, making it a promising material for advancing battery technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **A dipole-driven crystal sieve for crystal plane regulation via trace electrolyte additive towards stable and highly reversible zinc anodes** — score: 0.400 The article presents a novel method involving a dipole-driven crystal sieve enhanced by trace electrolyte additives to regulate crystal planes in zinc anodes. This approach aims to improve the stability and reversibility of zinc anodes, potentially addressing common issues associated with their performance in energy storage applications. Journal: *ScienceDirect Publication: Nano Energy*
- **Malleable textured Li for practical lithium metal batteries over 8,902 W kg−1** — score: 0.300 Wang et al. present a novel cryogenic rolling method to create malleable, textured lithium foils that improve mechanical stability and reduce ion diffusion barriers. This innovation effectively addresses key challenges associated with lithium metal anodes, paving the way for the development of high-power batteries suitable for electric aviation applications, achieving energy densities of over 8,902 W kg−1. Journal: *Joule*
- **Origin of Mn dissolution-triggered chain reactions in aqueous sodium-ion batteries** — score: 0.300 The article investigates the phenomenon of manganese dissolution in aqueous sodium-ion batteries and its role in initiating chain reactions that impact battery performance. The authors analyze the underlying mechanisms and implications for battery stability, providing insights that could enhance the design of more reliable energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Upcycling Spent Graphite via a Robust SiOx/C Coating for High-Capacity and Stable Lithium Storage** — score: 0.300 The article discusses a novel approach to upcycle spent graphite by applying a durable SiOx/C coating, which enhances its capacity and stability for lithium storage applications. This method potentially improves the performance and sustainability of lithium-ion batteries, addressing the challenges of waste graphite disposal. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **A Multi-Stage Modelling Framework for Accurate Early-Life SOH and EOL Prediction** — score: 0.300 The article presents a multi-stage modeling framework designed for precise early-life State of Health (SOH) and end-of-life (EOL) predictions of lithium-ion batteries, emphasizing the importance of accurate SOH estimation for optimal performance and safety. It utilizes functional principal component analysis (FPCA) to enhance predictive capabilities, showcasing innovative methodologies in battery health monitoring. Journal: *RSC - EES Batteries latest articles*

- **[ASAP] Ion-Trapping Binder for Ceramic-Coated Separators to Suppress Transition Metal Crossover in NCM811 Batteries** — score: 0.300 The article discusses a novel ion-trapping binder designed for ceramic-coated separators in NCM811 batteries, aiming to mitigate the crossover of transition metals, which can degrade battery performance. This innovative approach enhances the stability and efficiency of lithium-ion batteries by preventing unwanted metal ion migration. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Polyanionic Sodium Iron Pyrophosphate (NFP) at Low Sodium-to-Iron Ratio Incorporated with Carbon Nanofiber** — score: 0.300 The article explores the integration of polyanionic sodium iron pyrophosphate (NFP) with carbon nanofiber at a low sodium-to-iron ratio, highlighting its potential for improving energy storage materials. The study aims to enhance the electrochemical performance and stability of these materials for applications in batteries. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Solvent-confinement strategy realizes high-power anode-free sodium batteries** — score: 0.300 The article presents a novel solvent-confinement strategy that enhances the performance of anode-free sodium batteries, achieving high power output. The research highlights significant improvements in energy density and charge/discharge rates, marking a substantial advancement in battery technology for energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] A Binder-Free MOF-Based Separator Enabling Highly Selective Ion Sieving in Lithium-Ion Battery** — score: 0.300 The article presents a novel binder-free separator utilizing metal-organic frameworks (MOFs) that enhances ion selectivity in lithium-ion batteries. This innovative approach aims to improve battery efficiency by enabling precise ion sieving, thereby optimizing the overall performance of the energy storage system. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **Data-driven development of a universal quantitative metric for lithium-ion battery cathode rate performance** — score: 0.300 This article presents a new, comprehensive quantitative metric designed to evaluate the rate performance of lithium-ion battery cathodes, derived from extensive data analysis. The metric aims to standardize performance assessment across different materials, facilitating better comparisons and advancements in battery technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Composite air electrodes based on BaCe0.7Zr0.1Y0.1Yb0.1O3− for reversible solid oxide cells** — score: 0.200 The article discusses advancements in reversible solid oxide cells by combining a Gd0.3Ca2.7Co3.82Cu0.18O9− catalyst with a BaCe1−xZrxO3-based composite material. This integration improves the kinetics of reactions at the oxygen electrode, resulting in enhanced overall cell performance. Journal: *Nature Energy*
- **Ion-directed polymer/inorganic interphase with enriched charge for the stable and high-rate Zn anode** — score: 0.200 The article discusses the development of an ion-directed polymer/inorganic interphase designed to enhance the charge density at the Zn anode, addressing issues like concentration gradients and ion aggregation that impede performance. This innovation aims to improve the rate capability and stability of zinc-ion batteries, making them more viable for practical applications. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Conformal Transformation of Silicate to Si Nanotubes with Inward Volume Expansion for Fast-Charging Lithium Storage** — score: 0.200 The article discusses a novel method for transforming silicate materials into silicon nanotubes that undergo inward volume expansion, enhancing their performance for fast-charging lithium storage applications. The study highlights the promising implications of this transformation for improving energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Solvation Structure Engineering via Cyclic Amides in Aromatic Poly (oxindole biphenylene) Ion-Solvating Membranes for Stable Lithium Metal Batteries** — score: 0.200 The article discusses the development of ion-solvating membranes engineered with cyclic amides in aromatic poly(oxindole biphenylene) to enhance the stability of lithium metal batteries. This innovative ap-

proach aims to optimize the solvation structure, thereby improving battery performance and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Highly stable Mn-Sn flow batteries towards low-temperature energy storage** — score: 0.200 The article discusses the development of manganese-tin (Mn-Sn) flow batteries that demonstrate high stability and efficiency for energy storage at low temperatures, below the conventional threshold of $-20°C$. This advancement addresses challenges associated with the freezing point of electrolytes and low ionic conductivity in traditional aqueous redox flow batteries, potentially expanding their application in cold climates. Journal: *RSC - Energy Environ. Sci. latest articles*

- **Aqueous and non-aqueous aluminum metal anodes: Recent progress and comprehensive performance comparison** — score: 0.200 The article reviews recent advancements in aluminum metal anodes for energy storage, comparing their performance in both aqueous and non-aqueous environments. It provides a comprehensive analysis of the differences in efficiency, stability, and overall capabilities of these two types of systems, highlighting the implications for future battery technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Modulation of d-d orbital hybridization via creating asymmetrical paired iron sites for oxygen reduction reaction** — score: 0.200 This article discusses the development of diatomic catalysts with asymmetrical paired iron sites, which enhance the oxygen reduction reaction (ORR) by modifying d-d orbital hybridization. The authors propose a design principle aimed at optimizing the microenvironments of metal sites to improve catalytic activity, addressing the challenges posed by the sluggish kinetics of ORR. Journal: *RSC - Energy Environ. Sci. latest articles*